

REINE MIZERO

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Personal Summary

I am a Software Engineering student with experience in full-stack development, backend engineering, and machine learning. Skilled in JavaScript, TypeScript, Python, React, FastAPI, and TensorFlow, with experience building web applications, RESTful APIs, and AI-driven solutions. Interested in solving real-world problems in waste management, resource forecasting, and language technologies.

Education

African Leadership University

Sep 2023 - Present

- Bachelor of Science in Software Engineering
- Specialisation in Machine Learning, complemented by coursework in frontend and backend software development.

Maranyundo Girls School

Jan 2019 - Aug 2022

- High School Diploma
 - Mathematics, Physics and Computer Science
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Experience

AI Engineering Trainee, The Gym Rwanda x Ojemba x MaibornWolff

Mar 2026 - Present

Relevant Coursework: Artificial Intelligence Engineering

- Enrolled in an industry-led AI engineering programme with hands-on project work and practical implementation.
- Building a RESTful backend for a vacation planner application using FastAPI, implementing route handling, request validation, and API documentation.

Software Development Coach, The Gym Rwanda

Mar 2026 - Present

Relevant Responsibilities: Peer Coaching, Mentorship

- Mentor junior trainees in software development fundamentals, including JavaScript, React, and problem-solving techniques.
- Support structured learning by guiding students through project-based exercises and providing constructive feedback to accelerate their progress.

Software Developer Intern, Cylacon LTD

Jan 2026 - Apr 2026

Relevant Responsibilities: Web Developer, UI&UX Engineer

- Developed and maintained responsive web applications using React, Next.js, and modern UI practices to improve user experience and accessibility.
- Collaborated with cross-functional teams to design, implement, and document RESTful APIs and frontend features aligned with client requirements.

Software Development Trainee, The Gym Rwanda

Nov 2024 - Feb 2026

Relevant Coursework: Software Development

- Built full-stack applications using JavaScript, React, and Node.js while applying agile methodologies and version

control (Git).

- Strengthened professional skills, including communication, teamwork, incorporating feedback, and structured problem-solving, through project-based training.
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Technical Skills

Programming Languages: JavaScript, TypeScript, Python, Dart.

Frameworks & Libraries: React, NestJS, Node.js, Flutter, Tailwind CSS.

Databases & Backend Services: Firebase, MongoDB.

Tools & Platforms: Git, GitHub, REST API, Swagger UI, Render, Railway.

Machine Learning: Supervised learning, deep learning (TensorFlow), and large language model (LLM) applications.

Projects

Waste Classifier Model (MobileNetV2, Flask + TensorFlow, React, Railway (Cloud Platform), Locust)

<https://github.com/MizeroR/realwaste-classifier/>

- This project implements an end-to-end MLOps solution for automated waste classification using deep learning.
- The system classifies waste images into 9 categories to help improve waste sorting and recycling processes.

Kinyarwanda Courtesy Learning App (Prototype) (React, MBAZA TTS API)

<https://learnkinyarwanda.netlify.app/>

- Designed an educational app to help parents teach children polite expressions and greetings in Kinyarwanda.
- Features: interactive flashcards, quizzes, and text-to-speech integration.

Water Usage Forecasting ML Project (FastAPI, Flutter, ML Models)

<https://linear-regression-model-tqfq.onrender.com/docs>

- Built a machine learning model to predict water usage efficiency based on weather data.
 - Deployed the backend using FastAPI on Render.
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Volunteering Activities

The Gym Preparatory Training Program at Harambe Kimisagara Centre

Apr 2025 - Sep 2025

- Volunteered to train gap year students in basic web development skills (HTML, CSS, JavaScript).

Maran Charity at Maranyundo Girls School

Jan 2019 - Aug 2022

- Organised and supported fundraising initiatives to cover school fees for one primary school student per trimester.

Interests

French: Enjoy French literature, films, and podcasts.